

● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

01

Q1

ADVANCED MATH

A company has a newsletter. In January 2018, there were 1,300 customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c , subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

- A. $c = 1,3000.07^m$
- B. $c = 1,3001.07^m$
- C. $c = 1,3001.7^m$
- D. $c = 1,3007^m$

The population of the town of Smithville doubled every 75 years from 1659 to 1959. The population of this town was 240,000 in 1959. What was the population of this town in 1659?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

The function p is defined by $pn = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

- A. 2
- B. $\frac{8}{3}$
- C. 7
- D. 8

Q4

ADVANCED MATH

The function f is defined by $f(x) = |x - 4x|$. What value of a satisfies $f(5) - f(a) = -15$?

- A. -20
- B. 5
- C. 10
- D. 45

$$y = 3,600a^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y , in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A rubber ball bounces upward one-half the height that it falls each time it hits the ground. If the ball was originally dropped from a distance of 20.0 feet above the ground, what was its maximum height above the ground, in feet, between the third and fourth time it hit the ground?

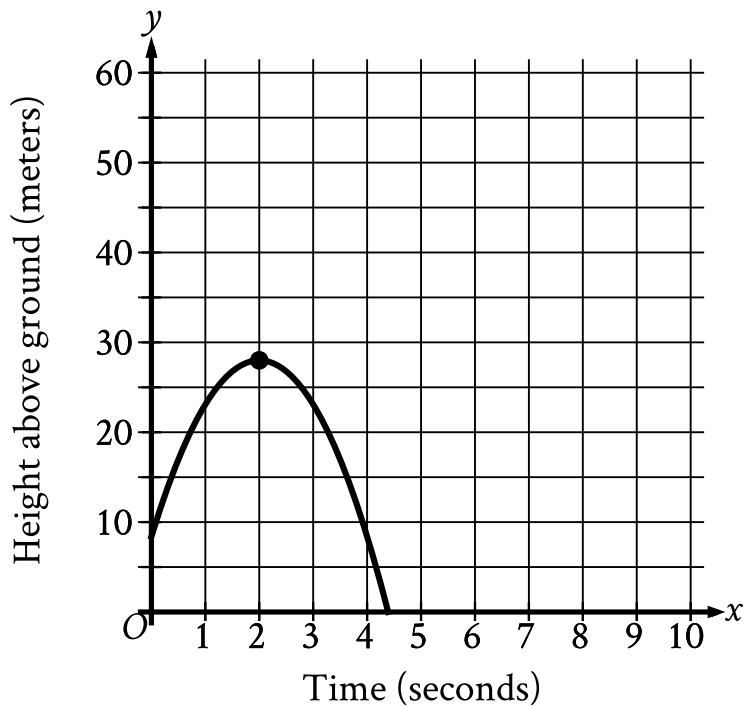
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s(87) = 23$ in this context?

- A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.
- B. The mass of the sample has decreased by approximately 23 grams 87 years after the sample starts to decay.
- C. The mass of the sample has decreased by approximately 87 grams 23 years after the sample starts to decay.
- D. Approximately 87 grams of the sample remains 23 years after the sample starts to decay.



- The parabola opens downward.
- The vertex is at point (2 comma 28).
- The parabola passes through the following points:
 - (0 comma 8.4)
 - (2 comma 28)
 - (4 comma 8.4)

An object was launched upward from a platform. The graph shown models the height above ground, y , in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

- A. From $x = 0$ to $x = 2$
- B. From $x = 0$ to $x = 4$

C. From $x = 2$ to $x = 3$

D. From $x = 3$ to $x = 4$

Q9

ADVANCED MATH

$$f(x) = 4,000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after 2012, where $x \leq 7$. Which of the following is the best interpretation of 4,000 in this context?

- A. The estimated initial beetle population for this species and area in 2012
- B. The estimated beetle population for this species and area 7 years after 2012
- C. The estimated percent decrease in the beetle population for this species and area each year after 2012
- D. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

The product of two positive integers is 546. If the first integer is 11 greater than twice the second integer, what is the smaller of the two integers?

- A. 7
- B. 14
- C. 39
- D. 78